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Opportunistic Relocation of Massive Coral Colonies in the Arabian Gulf: Survival Outcomes from a Coastal Development Mitigation Case Report

Running title: Opportunistic coral relocation in the Gulf

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ABSTRACT

Coral relocation is frequently used as a mitigation action when coastal construction affects reef habitat, yet applied outcomes are rarely reported for whole massive colonies in thermally extreme marginal seas. This case report documents an opportunistic whole colony relocation conducted during marine construction activities in the Saudi Arabian Gulf. Thirty-seven colonies belonging to *Dipsastraea*, *Platygyra*, and *Porites* were detached from a nearshore source reef, transported submerged in shaded aerated seawater, and reattached at a recipient site approximately 18 km away and 5 m deep. Colonies were fixed to either natural seabed or introduced concrete bases using underwater epoxy and were monitored after approximately 15 and 20 months. Overall survival after 20 months was 35.1%. Survival was 35.7% for *Dipsastraea*, 42.1% for *Platygyra*, and 0% for *Porites*. Final colony fate included 13 colonies alive, 20 dead in situ, and 4 missing/detached after 20 months. Exploratory logistic regression did not detect significant effects of genus or attachment substrate on survival, but inference was limited by the small sample size, uneven genus representation, complete mortality of *Porites*, and absence of control colonies. Missing and detached colonies were treated as non survivors because they were functionally lost from the mitigation site. This case provides a practical benchmark for evaluating coral relocation under real project constraints and identifies three management lessons: field conditions during relocation should be documented carefully, attachment design is a critical failure point, and post relocation monitoring should extend beyond one year to capture delayed mortality and mechanical loss.

Keywords: coral relocation, attachment failure, environmental management, impact mitigation, Arabian Gulf

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1. INTRODUCTION

Marine construction in coastal reef environments often generates a direct management question: whether coral colonies that cannot be avoided should be left to be lost, moved as part of mitigation, or excluded from mitigation plans because survival is uncertain. Coral relocation has therefore become a common applied response in reef impact management, but published outcomes remain uneven, especially for whole massive colonies moved under operational constraints rather than under controlled restoration trials.

The Arabian Gulf is a particularly severe setting for this type of intervention. Coral communities in the region persist under large seasonal temperature variation and recurrent thermal stress, while coastal development continues to affect shallow marine habitats (Riegl and Purkis, 2012; Burt, 2014; Lin et al., 2021). Coastal infrastructure development in the Saudi Arabian Gulf has highlighted the importance of integrating ecological considerations into engineering design and environmental monitoring, including the colonization of artificial substrates associated with offshore developments (Lozano-Cortés et al., 2019) and the management of sediment dispersion near sensitive marine habitats during construction activities (Saleem et al., 2026). The Gulf has also experienced repeated coral community changes associated with thermal extremes, coastal modification, and recurrent disturbance, making local documentation of mitigation outcomes especially important (Sheppard et al., 2010; Bento et al., 2016; Gower et al., 2025). Most regional restoration and relocation reports emphasize branching corals, fragments of opportunity, nursery-based approaches, or large relocation programs that are not always documented at the colony level (Sen and Yousif, 2016; Nithyanandan et al., 2018; Boström-Einarsson et al., 2020; Seraphim et al., 2024). Recent studies from the Arabian Gulf indicate a relative shift away from structurally complex branching coral assemblages following recurrent bleaching and thermal stress, with stress-tolerant massive and submassive taxa such as *Porites* and faviids becoming increasingly important components of some reef communities (Bento et al., 2016; Hadj-Hammou et al., 2025). This regional pattern is also consistent with recent Saudi Arabian Gulf studies documenting the apparent local disappearance of the formerly dominant branching coral *Stylophora pistillata* and sharp declines in *Acropora* spp. cover (Lozano-Cortés et al., 2025; Dixit et al., 2026). Whole massive colonies are therefore relevant for mitigation, but their relocation is logistically difficult and their survival trajectories are poorly documented.

This case report describes the field application and outcome of an opportunistic relocation of whole massive coral colonies during marine construction in eastern Saudi Arabia. The objective is not to test relocation effectiveness experimentally. Instead, the aim is to report a documented applied case, quantify survival and early growth outcomes, identify practical failure points, and extract management lessons that can inform future mitigation planning in thermally extreme coastal systems. Specifically, we quantified survival after 15 and 20 months, summarized outcomes by genus and attachment substrate, evaluated missing or detached colonies as a management relevant fate, and identified practical considerations for future coral relocation efforts in data limited reef systems.

2. MATERIALS AND METHODS

2.1 Case Description and Study Area

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The relocation was conducted as an applied mitigation action in response to impending coastal construction. Source colonies were collected from a nearshore reef at Abu Ali, eastern Saudi Arabia, at 27.137 °N and 49.667 °E, at about 5 m depth. The recipient site was located approximately 18 km northeast at 27.295 °N and 49.704 °E, at about 5 m depth. The recipient area included sandy patches and hard attachment points suitable for reattachment attempts.

This was not a planned experiment. All suitable massive colonies within the construction footprint that could be safely handled were moved to one recipient site. No unmanipulated control colonies were retained at the source reef, and no alternative depths, habitats, or attachment treatments were formally tested. Handheld field instruments recorded water temperatures of 32 to 33 °C at both source and recipient sites on the day of relocation. These measurements represent spot observations during the relocation operation and were not part of a continuous temperature monitoring program. Continuous temperature, turbidity, sedimentation, and hydrodynamic data were not collected after relocation.

2.2 Data Collection

On 23 August 2018, 37 whole massive coral colonies were detached using hammer and chisel. The transferred colonies included 14 *Dipsastraea*, 19 *Platygyra*, and 4 *Porites* colonies. Initial maximum colony diameters ranged from 10 to 25 cm. Mean initial diameter was 17.4 ± 4.3 cm for *Dipsastraea*, 18.5 ± 4.1 cm for *Platygyra*, and 16.5 ± 3.0 cm for *Porites*. Colonies were placed in plastic crates, lifted onto a vessel, and transported submerged in a shaded aerated seawater tank (Fig. 1). The relocation operation lasted approximately four hours, from 10:00 to 14:00 h, and no intentional air exposure occurred during transport.



Figure 1. Field sequence of the coral relocation process. Top: removal of whole coral colonies from the source reef. Middle: transport of colonies submerged in shaded seawater tanks. Bottom: reattachment of relocated colonies at the recipient site.

At the recipient site, attachment surfaces were cleaned with metal brushes to remove loose sediment and surface biota. Colonies were attached using underwater epoxy either to natural seabed or to introduced concrete bases. The concrete bases were introduced as attachment structures during the relocation operation and were not preexisting natural features of the recipient reef. The relocated colonies were tagged, spaced approximately 1 m apart, and photographed after attachment. Colony allocation to

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attachment substrate was operational and not randomized. Overall, 18 colonies were attached to concrete bases and 19 colonies were attached to natural substrate. Within genera, 7 *Dipsastraea* colonies were attached to concrete and 7 to natural substrate, 9 *Platygyra* colonies were attached to concrete and 10 to natural substrate, and 2 *Porites* colonies were attached to concrete and 2 to natural substrate. Initial colony diameter and substrate allocation are summarized in Supplementary Table S1.

Monitoring surveys were conducted on 14 November 2019 and 16 April 2020, approximately 15 and 20 months after relocation. During each survey, colonies were visually classified as alive, dead in situ, or missing/detached (Fig. 2). Colonies classified as dead in situ were still present at the relocation site but showed no visible live tissue. Colonies classified as missing/detached were not found attached at their original relocation position, or the attachment base was present but the relocated colony was absent. Colonies that detached and could not be located near their original position were classified as non survivors because they were functionally lost from the mitigation site. Radial extension was measured only for colonies that were alive at the final monitoring event and refers to the observed increase in live colony radius between relocation and the final survey. Dead and missing/detached colonies were not included in radial extension summaries.

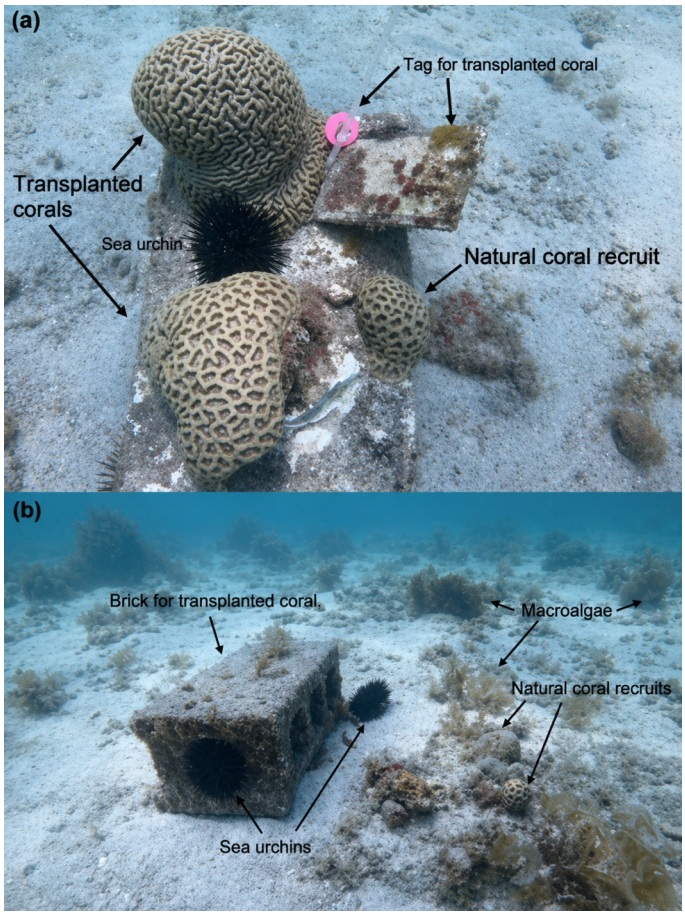


Figure 2. Field photos from the final monitoring survey on 16 April 2020. (a) Surviving relocated colony. (b) Introduced concrete base where the relocated colony was missing/detached. Natural coral recruits, sea urchins, and macroalgae were also observed around the relocation site.

2.3 Statistical Analysis

Survival at 20 months was summarized by genus and attachment substrate. Because this relocation was an opportunistic mitigation action, all statistical analyses were exploratory and were interpreted cautiously. Initial colony diameter was summarized by genus using mean, standard deviation, and range, and differences among genera were evaluated using a Kruskal-Wallis test. Associations between survival and attachment substrate were evaluated using

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Fisher's exact test. Associations between survival and genus were evaluated using a Fisher-Freeman-Halton exact test, which is appropriate for contingency tables with small expected counts.

Final colony fate at 20 months was summarized by attachment substrate, separating colonies that were alive, dead in situ, and missing/detached. The association between missing/detached status and attachment substrate was evaluated using Fisher's exact test. To evaluate whether final radial extension was related to initial colony size, the correlation between initial colony diameter and radial extension among surviving colonies was tested using Spearman's rank correlation.

A logistic regression was used to explore whether genus or substrate predicted final survival, but this analysis was interpreted cautiously because the sample size was small and the field action lacked experimental controls. Because no *Porites* colonies survived, this genus could not be reliably included in the logistic regression model. We therefore report the descriptive outcome for all genera but used a reduced logistic regression including only *Dipsastraea* and *Platygyra* to estimate the effects of genus and attachment substrate on survival. The model results are interpreted as exploratory rather than confirmatory because colony allocation was not randomized and sample size was limited.

3. RESULTS

Nineteen of the 37 relocated colonies were alive at the 15-month survey, equivalent to 51.4% survival. Thirteen of the 37 relocated colonies survived to the 20-month survey, equivalent to 35.1% overall survival. Survival at 20 months was reported descriptively by genus and was not statistically associated with genus (Fisher-Freeman-Halton exact test, $p = 0.384$). Five of 14 *Dipsastraea* colonies survived, equivalent to 35.7%. Eight of 19 *Platygyra* colonies survived, equivalent to 42.1%. None of the four *Porites* colonies survived (Fig. 3a). At 15 months, survival was 8 of 14 colonies for *Dipsastraea* (57.1%), 11 of 19 for *Platygyra* (57.9%), and 0 of 4 for *Porites* (0%). Initial colony diameter did not differ significantly among genera (Kruskal Wallis test, $H = 1.01$, $p = 0.602$).

Mean radial extension among surviving colonies was 0.83 ± 0.38 cm for *Dipsastraea* and 1.55 ± 0.63 cm for *Platygyra*. No radial extension was reported for *Porites* because no colonies survived to the final monitoring survey. Across all surviving colonies, radial extension averaged 1.28 ± 0.64 cm and was not correlated with initial colony diameter (Spearman rho $= -0.072$, $p = 0.815$).

Survival at 20 months was similar between attachment substrates. Seven of 18 colonies attached to concrete bases survived (38.9%), compared with 6 of 19 colonies attached to natural substrate (31.6%). Survival was not significantly associated with attachment substrate (Fisher's exact test, $p = 0.737$). Final colony fate at 20 months included 13 colonies alive (35.1%), 20 dead in situ (54.1%), and 4 missing/detached (10.8%). Among colonies attached to concrete bases, 7 were alive (38.9%), 9 were dead in situ (50.0%), and 2 were missing/detached (11.1%). Among colonies attached to natural substrate, 6 were alive (31.6%), 11 were dead in situ (57.9%), and 2 were missing/detached (10.5%) (Fig. 3b). Missing/detached status was not significantly associated with attachment substrate (Fisher's exact test, $p = 1.000$).

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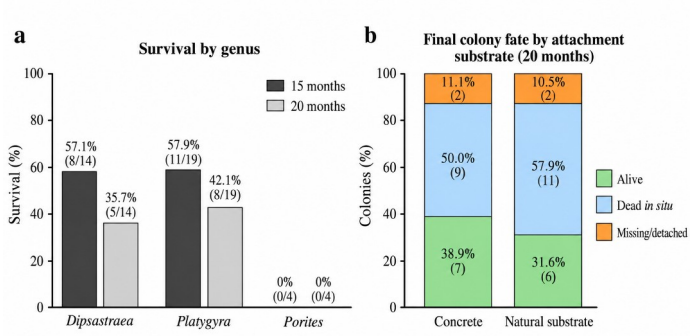


Figure 3. Survival and final fate of relocated coral colonies. (a) Survival rate (%) by genus at 15 and 20 months after relocation. Labels above bars indicate the number of surviving colonies relative to the number relocated. (b) Final colony fate by attachment substrate at 20 months, separating colonies that were alive, dead in situ, or missing/detached.

The logistic regression did not detect a statistically significant effect of genus or attachment substrate on survival. The full model including all genera was unstable because no *Porites* colonies survived, producing quasi complete separation. A reduced logistic regression including only *Dipsastraea* and *Platygyra* also did not detect statistically significant effects of genus or attachment substrate on survival. In this reduced model, the odds ratio for *Platygyra* relative to *Dipsastraea* was 1.32 (95% CI: 0.32 to 5.53, p = 0.700), and the odds ratio for natural substrate relative to concrete was 0.70 (95% CI: 0.17 to 2.84, p = 0.612). The likelihood ratio test for the reduced model was not significant (p = 0.820). Exploratory statistical analyses, including the reduced logistic regression results, are summarized in Supplementary Table S2. This result should not be read as evidence that taxonomic identity or attachment design are unimportant. The dataset was too small to detect moderate effects with confidence, and the lack of replication among recipient sites further limits inference. The observed complete loss of *Porites* and the contribution of detachment or disappearance to non-survival remain management relevant even if formal statistical separation was not possible.

4. DISCUSSION

4.1 Scientific and Management Value of the Case

The main value of this case is that it documents a real mitigation action under operational constraints rather than an optimized restoration trial. Such cases are often implemented by project teams but remain unpublished, which creates a bias toward better documented success stories and limits realistic planning for coral relocation and restoration actions (Boström-Einarsson et al., 2020; Seraphim et al., 2024). The 35.1% survival reported here is modest, but it provides a useful benchmark for future environmental impact assessments, permit conditions, and monitoring plans involving whole colony transfer in thermally stressful coastal systems, including the Arabian Gulf, where coastal development and environmental extremes place strong pressure on reef habitats (Burt, 2014; Riegl and Purkis, 2012; Sheppard et al., 2010).

The case points to three practical lessons. First, timing should be considered in relocation planning, although it was not directly tested in this study. The relocation took place in August, when water temperatures measured during the operation were 32 to 33 °C. Because continuous temperature data were not collected after relocation, we cannot determine whether thermal stress contributed directly to the observed mortality or whether the relocation coincided with a marine heatwave. Future projects should avoid the hottest and coldest seasonal windows where schedules allow, because relocation stress may interact with thermal stress in thermally extreme reef systems. Second, attachment stability

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should be treated as a core design component rather than a minor field detail. Missing or detached colonies represent functional mitigation failure even if tissue mortality cannot be confirmed. In this case, 10.8% of colonies were missing or detached at the final survey, with similar proportions on concrete bases and natural substrate (Fig. 3b). Third, monitoring should extend beyond a single post relocation visit. The difference between the 15-month and 20-month observations indicates that delayed losses can occur after the initial establishment period and that short monitoring windows may overestimate relocation success. The case also has scientific value because it reports colony level outcomes for massive coral genera that are less represented in published relocation datasets than branching taxa.

The complete loss of *Porites* colonies is notable but should be interpreted cautiously because only four colonies of this genus were relocated. Although *Porites* is often considered environmentally tolerant and can persist under challenging conditions outside typical reef settings, including estuarine habitats (Zapata and Lozano-Cortés, 2015), the result observed here does not demonstrate genus specific vulnerability, because mortality may also have reflected handling stress, attachment failure, local site conditions, or the small and unbalanced sample size. The complete loss of *Porites* should therefore be treated as a management relevant observation that requires testing with larger and more balanced relocation datasets. Similarly, the absence of statistically significant effects of genus or attachment substrate should not be interpreted as evidence that these factors are unimportant. Rather, it reflects the limited inferential power of a small, unreplicated, opportunistic mitigation dataset.

4.2 Limitations of Inference

This case report should not be interpreted as a controlled test of coral relocation effectiveness. The action involved one source area, one recipient site, a small number of colonies, three genera, and no unmanipulated controls. Attachment substrate was not randomly assigned, and colonies were not distributed across multiple recipient habitats or depths. Environmental drivers such as turbidity, sedimentation, hydrodynamic exposure, and thermal variability were not monitored continuously. Mortality in the analysis also includes missing colonies that may have detached or been displaced. These limitations prevent attribution of losses to a single cause.

The statistical analyses should therefore be interpreted as exploratory summaries rather than confirmatory tests. Exact tests and logistic regression were useful for making the interpretation more transparent, but the small and uneven sample sizes limited statistical power. The absence of surviving *Porites* colonies also produced instability in the full logistic model, requiring a reduced model focused on *Dipsastraea* and *Platygyra*. These constraints reinforce the need to treat the results as a documented mitigation outcome rather than as a formal test of genus or substrate effects.

The practical interpretation is therefore narrow but useful. The case shows that whole colony transfer of massive corals is technically feasible under project conditions in the Saudi Arabian Gulf, but it also shows that survival can be low when relocation is opportunistic, attachment is difficult, and work occurs in an environmentally extreme system. Future applications should be planned as adaptive mitigation actions with documented baselines, stronger

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attachment designs, clear monitoring schedules, continuous environmental monitoring where feasible, and explicit decision thresholds for success or failure.

5. CONCLUSIONS

This case report documents survival outcomes from an opportunistic relocation of massive coral colonies during marine construction in the Saudi Arabian Gulf. The outcome was mixed: relocation was technically possible, but only 35.1% of colonies survived after 20 months, and detachment or disappearance contributed to functional losses. The decline in survival between the 15- and 20-month surveys highlights the importance of monitoring beyond an initial post relocation assessment. The case supports a cautious management position.

Whole colony relocation should not be treated as a guaranteed offset for reef impacts. It can be considered where impact avoidance is not possible, but it should be paired with careful seasonal planning where project schedules allow, robust attachment design, longer monitoring, quantitative reporting of missing or detached colonies, and transparent reporting of both survival and failure.

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CONFLICT OF INTEREST

The study was funded by Saudi Aramco, which employs three of the authors.

DATA AVAILABILITY STATEMENT

The colony level dataset supporting this case report is publicly available from figshare: https://doi.org/10.6084/m9.figshare.31939239.v1.

The dataset includes colony identity, genus, attachment substrate, initial diameter, survival status at approximately 15 and 20 months, radial extension for surviving colonies, and field notes.

AUTHOR CONTRIBUTIONS

D.L.C.: Conceptualization; Methodology; Data curation; Formal Analysis; Writing original draft; Writing review and editing.

R.v.d.M.: Writing review and editing.

A.Q.: Funding acquisition; Supervision; Project administration.

K.M., S.D.: Investigation; Data curation; Writing review and editing.

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SUPPLEMENTARY MATERIAL

Supplementary Table S1. Relocation design, initial colony size, survival, and final colony fate of relocated coral colonies by genus and attachment substrate.

Supplementary Table S2. Summary of exploratory statistical analyses used to evaluate initial colony diameter, survival, missing/detached status, radial extension, and logistic regression results.

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